



Student Name: _____

Teacher: _____

Sydney Technical High School

2024

HIGHER
SCHOOL
CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless work or illegible writing
- **Begin each question on a new page**

**Total marks:
100**

Section I — 10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this part

Section II — 90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours and 15 minutes for this section.

SECTION I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Answer each question on the multiple-choice page in your answer booklet.

1 Which one of the following is i^{2019} simplified?

- A. i
- B. $-i$
- C. 1
- D. -1

2 Consider the expansion of

$$(11x - 8)^{11} = \sum_{k=0}^{11} t_k x^k.$$

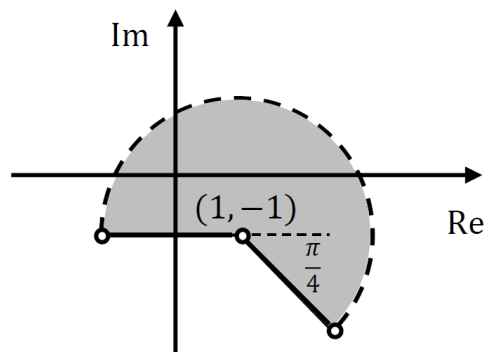
Which expression represents t_k ?

- A. $t_k = {}^{11}C_k(11x)^k(-8)^{11-k}$
- B. $t_k = {}^{11}C_k(11)^{11-k}(-8)^k$
- C. $t_k = {}^{11}C_k(11)^k(-8)^{11-k}$
- D. $t_k = {}^{11}C_k(11x)^k(-8)^k$

- 3 A particle undergoing simple harmonic motion in a straight line has an acceleration given by $\ddot{x} = 75 - 25x$, where x is the displacement after t seconds. Where is the centre of the motion?

- A. $x = 0$
- B. $x = 3$
- C. $x = 5$
- D. $x = 6$

- 4 The shaded region below is constructed by taking the intersections of two other regions.



Which of the following best represents the two possible regions?

- A. $|z - 1 + i| < 2$ and $-\frac{\pi}{4} \leq \arg(z - 1 + i) \leq \pi$
- B. $|z - 1 + i| \leq 2$ and $-\frac{\pi}{4} < \arg(z - 1 + i) < \pi$
- C. $|z + 1 - i| < 2$ and $-\frac{\pi}{4} \leq \arg(z + 1 - i) \leq \pi$
- D. $|z + 1 - i| \leq 2$ and $-\frac{\pi}{4} < \arg(z + 1 - i) < \pi$

5 Which of the following lines is parallel to $x = 3 + 2t, y = 2 - t, z = 3 + 4t$?

A. $\frac{x-2}{5} = \frac{y-1}{2} = \frac{z-5}{3}$

B. $\frac{x-2}{1} = \frac{y-1}{3} = \frac{z-5}{-1}$

C. $\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-5}{4}$

D. $\frac{x-2}{5} = \frac{y-1}{1} = \frac{z-5}{7}$

6 It is given that $2 - i$ is a root of $P(z) = z^3 + az^2 + bz + 20$ where a and b are real numbers. Factorize $P(z)$ over the real numbers.

A. $P(z) = (z + 4)(z^2 + 4z + 5)$

B. $P(z) = (z + 4)(z^2 - 4z - 5)$

C. $P(z) = (z + 4)(z^2 - 4z + 5)$

D. $P(z) = (z - 4)(z^2 - 4z + 5)$

7 Consider the following statement.

“If it is raining, then there are grey clouds in the sky.”

Which of the following is the contrapositive of this statement?

- A. If there are grey clouds in the sky, then it is raining.
- B. If it is not raining, then there are no grey clouds in the sky.
- C. If it is raining, then there are no grey clouds in the sky.
- D. If there are no grey clouds in the sky, then it is not raining.

8 At time t , the position vector of a particle is given by $\vec{r}(t) = \sqrt{t}\vec{i} + \frac{t-1}{t}\vec{j}$. With what equation does this particle moves along the path?

- A. $y(x^2 + 1) = 1$
- B. $x(y^2 - 1) = 1$
- C. $y = x^2 - 1$
- D. $y = \frac{x^2-1}{x^2}$

- 9 Consider the following definite integrals

$$I_1 = \int_2^1 \frac{1}{\sqrt{4-x^2}} dx$$

$$I_2 = \int_0^1 \cos(\cos^{-1} x) dx$$

$$I_3 = \int_0^1 \sin^{-1} x dx$$

$$I_4 = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \cos(x^2) \cos(2x^2) dx$$

Which of the following correctly lists the values of these definite integrals in **ascending** order?

- A. I_1, I_2, I_3, I_4
- B. I_4, I_2, I_3, I_1
- C. I_1, I_4, I_3, I_2
- D. I_1, I_4, I_2, I_3

- 10 $f(x)$ is any continuous and differentiable functions over $\mathbb{R}$.

Given $2 + 3f(x) = f(-x) + 3 \int_{-1}^1 f(x) dx$, what is the value of $\int_{-1}^1 f(x) dx$?

- A. 2
- B. 1
- C. -1
- D. -2

SECTION II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in your answer booklet. Start each question on a new page.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) START A NEW PAGE

(a) If $z = 3 - i$ and $w = 1 + 2i$, find in the form $a + ib$, where a and b are real, the values of

(i) $z - 2w$ 1

(ii) $z\overline{w}$ 1

(iii) $\frac{z}{w}$ 1

(b) Find $\int \frac{x}{1+x^2} dx$ 1

(c) (i) Express $-2 + 2i$ in exponential form. 2

(ii) Hence, simplify $(-2 + 2i)^{8k}$, where k is an integer. 2

Question 11 continues on page 9

Question 11 (continued)

- (d) A particle is experiencing simple harmonic motion along a straight line. At time t seconds, its displacement x metres from a fixed point O on the line is given by

$$x = 8 \cos^2 t + 1$$

- (i) Show that $\ddot{x} = -4(x - 5)$. **2**
- (ii) Find the centre and period of motion. **2**
- (e) Evaluate $\int_3^4 x\sqrt{x-3}dx$. **3**

End of Question 11

Question 12 (16 marks) START A NEW PAGE

- (a) Find the size of the angle between the vectors $\begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 2 \\ 5 \end{pmatrix}$, correct to the nearest degree. **2**
- (b) Prove the point $\begin{pmatrix} 8 \\ 12 \\ -16 \end{pmatrix}$ does not lie on $\tilde{r} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix}$ **2**
- (c) (i) Split $\frac{8}{(x+2)(x^2+4)}$ into partial fractions. **2**
- (ii) Hence, evaluate $\int_0^2 \frac{8}{(x+2)(x^2+4)} dx$. **3**
- (d) Find the intersections of the line $\tilde{r} = 3\tilde{i} + 2\tilde{j} - \tilde{k} + \lambda(2\tilde{i} - \tilde{j} - 2\tilde{k})$ with the sphere $(x-3)^2 + (y-2)^2 + (z+1)^2 = 36$. **3**
- (e) On the Argand diagram, vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the complex numbers $z_1 = 2\left(\cos\frac{4\pi}{5} + i\sin\frac{4\pi}{5}\right)$ and $z_2 = 2\left(\cos\frac{7\pi}{15} + i\sin\frac{7\pi}{15}\right)$ respectively.
- (i) Show that $\triangle OAB$ is equilateral. **2**
- (ii) Express $z_2 - z_1$ in *exact* modulus-argument form. **2**

End of Question 12

Question 13 (14 marks) START A NEW PAGE

(a) A particle is moving in simple harmonic motion in a straight line and its velocity is given by $v^2 = 6 - x - x^2$ m/s, where x is the displacement in metres.

(i) Find the two points between which the particle is oscillating. **1**

(ii) Find the centre of motion. **1**

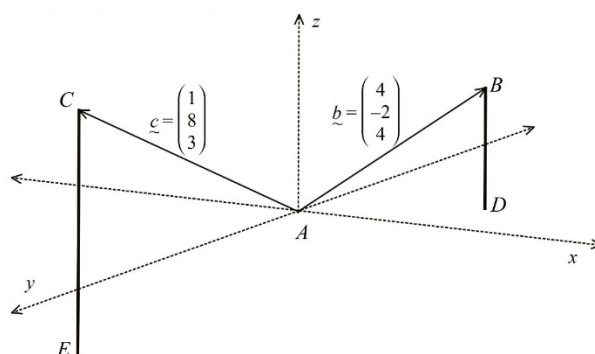
(iii) Find the maximum speed of the particle. **1**

(iv) Find the acceleration of the particle in terms of x . **1**

(b) Use the substitution $t = \tan x$ to find $\int \frac{dx}{1 + \sin 2x}$ **3**

(c) Two vertical posts CE and BD stand on a horizontal plane ADE . **2**

Using A as the origin, the vectors $\vec{b}$ and $\vec{c}$ represent the locations of the top of each post, B and C .



Show that $\triangle ABC$ is right angled at A .

Question 13 continues on page 12

Question 13 (continued)

(d) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$, where n is a non-negative number.

(i) Show that $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$, when $n \geq 2$. **2**

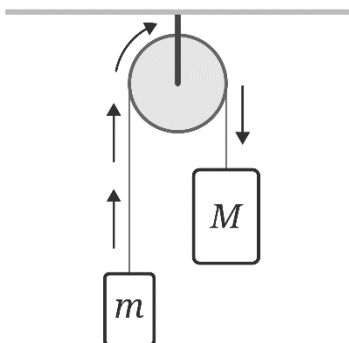
(ii) Deduce that $I_n = \frac{(n-1)}{n} I_{n-2}$, when $n \geq 2$. **2**

(iii) Evaluate I_4 . **1**

End of Question 13

Question 14 (14 marks) START A NEW PAGE

- (a) Box m has a mass of 4 kg and Box M has a mass of 6 kg. The boxes are connected by a light inextensible string passing over a frictionless pulley. Initially, the boxes are at rest. After Box m has travelled x metres in an upwards direction, it is travelling at v metres per second. **3**



Prove that $v = \sqrt{\frac{2gx}{5}}$

- (b) Use mathematical induction to prove that **3**

$$\frac{d^n}{dx^n}(xe^{2x}) = 2^{n-1}(2x + n)e^{2x}$$

for $n \geq 1$.

Question 14 continues on page 14

Question 14 (continued)

- (c) A mass of 1 kg is falling under gravity (g) through a medium in which the resistance to the motion is proportional to the square of the velocity.

Let k be the constant of proportionality.

- (i) Write an equation for the acceleration of this mass. **1**

- (ii) Show that the mass reaches a terminal velocity given by $v = \sqrt{\frac{g}{k}}$. **1**

- (iii) Show that the distance it has fallen when it reaches a velocity v m/s is given **3**
by

$$x = \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$$

- (d) Prove by contradiction, or otherwise, that for $a \geq 2$; **3**

$$\sqrt{a} + \sqrt{a+2} > \sqrt{a+8}.$$

End of Question 14

Question 15 (15 marks) START A NEW PAGE

- (a) (i) Show that 1, 2 and 3 are the only positive integers that satisfy **1**

$$a^2 + b^2 + c^2 = 14$$

- (ii) $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$ are the position vectors of the points at the ends of a diameter of a sphere.

- (α) Find the centre and radius of the sphere. **2**

- (β) Determine how many points, with integer coefficients, lie on the sphere (include the two given points). **2**

- (b) (i) Prove that **2**

$$\int_{\frac{1}{a}}^a \frac{f(x)}{x \left(f(x) + f\left(\frac{1}{x}\right) \right)} dx = \int_{\frac{1}{a}}^a \frac{f\left(\frac{1}{x}\right)}{x \left(f(x) + f\left(\frac{1}{x}\right) \right)} dx$$

- (ii) Hence evaluate **3**

$$I = \int_{\frac{1}{2}}^2 \frac{\sin x}{x \left(\sin x + \sin \frac{1}{x} \right)} dx$$

Question 15 continues on page 16

Question 15 (continued)

(c) Let n be an integer greater than 2. Suppose ω is an n th root of unity and $\omega \neq 1$.

(i) By expanding the left-hand side, show that **2**

$$(1 + 2\omega + 3\omega^2 + 4\omega^3 + \cdots + n\omega^{n-1})(\omega - 1) = n$$

(ii) Using the identity $\frac{1}{z^2-1} = \frac{z^{-1}}{z-z^{-1}}$, or otherwise, prove that **1**

$$\frac{1}{\cos 2\theta + i \sin 2\theta - 1} = \frac{\cos \theta - i \sin \theta}{2i \sin \theta}$$

provided that $\sin \theta \neq 0$.

(iii) Hence, if $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$, find the real part of $\frac{1}{\omega-1}$. **1**

(iv) Deduce that **1**

$$1 + 2 \cos \frac{2\pi}{5} + 3 \cos \frac{4\pi}{5} + 4 \cos \frac{6\pi}{5} + 5 \cos \frac{8\pi}{5} = -\frac{5}{2}$$

End of Question 15

Question 16 (16 marks) START A NEW PAGE

(a) α and β are the roots of $x^2 - 2x + 4 = 0$.

(i) Prove that **3**

$$\alpha^n - \beta^n = i2^{n+1} \sin\left(\frac{n\pi}{3}\right)$$

(ii) Hence, find the value of $\alpha^9 - \beta^9$. **1**

(b) (i) Use de Moivre's theorem to show that **3**

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

(ii) Find the general solution of $\tan 4\theta = 1$. **1**

(iii) Hence find the roots of the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ in **2**
trigonometrical form.

(c) It is known that the arithmetic mean of three numbers is greater than the
geometric mean of those numbers, that is

$$\frac{a + b + c}{3} \geq \sqrt[3]{abc}$$

(i) Given that $a + b + c = t$ and $(a, b, c > 0)$, prove that **3**

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{t}$$

(ii) Hence, given $a + b + c = 1$ and $(a, b, c > 0)$, prove that **3**

$$\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \geq 8$$

End of paper

STHS Ext 2 Trial Solutions.

$$\begin{aligned} \textcircled{1} \quad i^{2019} &= i^{2019-2016} \\ &= i^3 \\ &= i^2 \times i \\ &= -1 \times i \\ &= -i \end{aligned}$$

B.

$$\begin{aligned} \textcircled{2} \quad (11x-8)^{11} &= \sum_{k=0}^{11} {}^{11}C_k (11x)^k (-8)^{11-k} \\ &= \sum_{k=0}^{11} {}^{11}C_k 11^k (-8)^{11-k} x^k \end{aligned}$$

$$t^k = {}^{11}C_k 11^k (-8)^{11-k}$$

C.

$$\begin{aligned} \textcircled{3} \quad \ddot{x} &= 75 - 25x \\ &= -25(x-3) \\ &= -5^2(x-3) \end{aligned}$$

$x=3$ is centre of motion B.

④ Circle centre $(1, -1)$ and radius is 2.

$$|z - (1-i)| < 2$$

$$|z - 1 + i| < 2$$

$$-\frac{\pi}{4} \leq \arg(z - 1 + i) \leq \pi$$

A.

$$\textcircled{5} \quad \underline{r} = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

↑

compare coefficients with
denominators

C.

$$\textcircled{6} \quad P(z) = z^3 + az^2 + bz + 20$$

Let roots be $2 \pm i$ and α .

$$(2+i)(2-i)\alpha = -\frac{20}{1}$$

$$(4-i^2)\alpha = -20$$

$$5\alpha = -20$$

$$\alpha = -4$$

$$P(z) = (z - -4)(z - (2+i))(z - (2-i))$$

$$= (z+4)(z-2-i)(z-2+i)$$

$$= (z+4)(z^2 - 4z + 5) \quad \text{C.}$$

$\textcircled{7}$ Contrapositive for: If P then Q
is: If not Q then not P. D.

$$\textcircled{8} \quad \begin{aligned} x &= \sqrt{t} \\ x^2 &= t \end{aligned}$$

$$y = \frac{t-1}{t}$$

$$y = \frac{x^2-1}{x^2}$$

D.

$$\textcircled{9} \quad I_1 < 0$$

$$I_2 = \int_{-\frac{1}{2}}^1 x dx$$

$$I_3 = \left(\frac{\pi}{2} \times 1\right) - \int_0^{\frac{\pi}{2}} \sin x dx \quad (\text{using areas})$$

$$= \frac{\pi}{2} - 1$$

$$= 0.5707...$$

$$I_4 = 0 \quad (\text{odd } f^n)$$

$$I_1, I_4, I_2, I_3$$

D.

$$(10) \quad 2 + 3f(x) = f(-x) + 3 \int_{-1}^1 f(x) dx.$$

$$\text{Let } I = \int_{-1}^1 f(x) dx.$$

$$2 + 3f(x) = f(-x) + 3I.$$

$$2 \int_{-1}^1 dx + 3 \int_{-1}^1 f(x) dx = \int_{-1}^1 f(-x) dx + 3I \int_{-1}^1 dx.$$

$$2[x]_{-1}^1 + 3I = \underbrace{\int_{-1}^1 f(-x) dx}_{\text{Let } u = -x} + 3I [x]_{-1}^1$$

$$\begin{aligned} &\rightarrow \text{Let } u = -x \\ &\quad -du = dx. \end{aligned}$$

$$x = -1, u = 1$$

$$x = 1, u = -1$$

$$4 + 3I = \int_1^{-1} -f(u) du + 6I$$

$$4 = \int_{-1}^1 f(u) du + 3I$$

$$4 = I + 3I$$

$$I = 1.$$

B.

$$\textcircled{11} \text{ a) } z = 3 - i, w = 1 + 2i$$

$$\text{(i) } z - 2w = 3 - i - 2(1 + 2i)$$

$$= 3 - i - 2 - 4i$$

$$= 1 - 5i$$

$$\text{(ii) } z\bar{w} = (3 - i)(1 - 2i)$$

$$= 3 - 7i + 2i^2$$

$$= 1 - 7i$$

$$\text{(iii) } \frac{z}{w} = \frac{3 - i}{1 + 2i} \times \frac{1 - 2i}{1 - 2i}$$

$$= \frac{3 - 7i + 2i^2}{1 - 4i^2}$$

$$= \frac{1 - 7i}{5}$$

$$\text{b) } \int \frac{x}{1+x^2} dx = \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

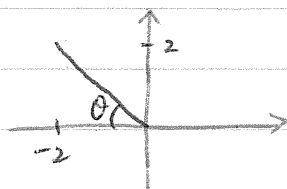
$$= \frac{1}{2} \ln|1+x^2| + C$$

$$\text{c) (i) } -2 + 2i$$

$$r = |-2 + 2i|$$

$$= \sqrt{2^2 + 2^2}$$

$$= 2\sqrt{2}$$



$$\theta = \pi - \tan^{-1}\left(\frac{2}{2}\right)$$

$$= \frac{3\pi}{4}$$

$$-2 + 2i = 2\sqrt{2} e^{i\frac{3\pi}{4}}$$

$$\text{(ii) } \left(2\sqrt{2} e^{\frac{3\pi i}{4}}\right)^{8k} = \left(2^{\frac{3}{2}}\right)^{8k} e^{\frac{3\pi i}{4} \times 8k}$$

$$= 2^{12k} e^{6k\pi i}$$

⑪ d) $x = 8\cos^2 t + 1$

(i) $x = 8\left(\frac{1}{2} + \frac{1}{2}\cos 2t\right) + 1$

$= 4 + 4\cos 2t + 1$

$= 4\cos 2t + 5 \rightarrow x - 5 = 4\cos 2t$

$\dot{x} = 4 \times 2 \times -\sin 2t$

$= -8\sin 2t$

$\ddot{x} = -8 \times 2\cos 2t$

$= -4 \times 4\cos 2t$

$= -4(x - 5)$

(ii) $C = 5$

$T = \frac{2\pi}{2}$

$= \pi \text{ seconds.}$

e) $\int_3^4 x \sqrt{x-3} \, dx$

$= \int_0^1 (u+3) u^{\frac{1}{2}} \, du$

$= \int_0^1 \left(u^{\frac{3}{2}} + 3u^{\frac{1}{2}}\right) \, du$

$= \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + 3 \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1$

$= \frac{2}{5} + 3\left(\frac{2}{3}\right)$

$= \frac{12}{5}$

Let $u = x - 3$
 $du = dx$

$$\textcircled{12} \text{ a) } \cos \theta = \frac{5(-3) + 3(2) + 2(5)}{\sqrt{5^2 + 3^2 + 2^2} \times \sqrt{(-3)^2 + 2^2 + 5^2}}$$

$$= \frac{1}{38}$$

$$\theta = 88^\circ$$

$$\text{b) } \begin{pmatrix} 8 \\ 12 \\ -16 \end{pmatrix} = \begin{pmatrix} -1 + 3\lambda \\ 0 + 4\lambda \\ 1 + 5\lambda \end{pmatrix} \quad \begin{array}{l} 8 = -1 + 3\lambda, \lambda = 3 \\ 12 = 4\lambda, \lambda = 3 \\ -16 = 1 + 5\lambda, \lambda = \frac{17}{5} \end{array}$$

$\therefore$ the point does not lie on r .

$$\text{c) (i) } \frac{8}{(x+2)(x^2+4)} = \frac{a}{x+2} + \frac{bx+c}{x^2+4}$$

$$8 = a(x^2+4) + (bx+c)(x+2)$$

$$8 = ax^2 + 4a + bx^2 + cx + 2bx + 2c$$

$$8 = (a+b)x^2 + (2b+c)x + (4a+2c)$$

$$a+b=0$$

$$a = -b \quad - \textcircled{1}$$

$$2b+c=0$$

$$2(-a)+c=0 \quad \text{using } \textcircled{1}$$

$$-2a+c=0 \quad - \textcircled{2}$$

$$4a+2c=8 \quad - \textcircled{3}$$

$$4a - 2(2a) + 2c - 2(c) = 8 - 2(0)$$

$$8a = 8$$

$$a = 1$$

$$b = -1$$

$$c = 2$$

$$\frac{8}{(x+2)(x^2+4)} = \frac{1}{x+2} + \frac{2-x}{x^2+4}$$

$$\begin{aligned}
 (12) \text{ c) (ii) } & \int_0^2 \left(\frac{1}{x+2} + \frac{2-x}{x^2+4} \right) dx \\
 &= \int_0^2 \frac{1}{x+2} dx + \int_0^2 \frac{2}{x^2+4} dx - \frac{1}{2} \int_0^2 \frac{2x}{x^2+4} dx \\
 &= \left[\ln|x+2| + 2 \times \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) - \frac{1}{2} \ln(x^2+4) \right]_0^2 \\
 &= (\ln 4 + \tan^{-1} 1 - \frac{1}{2} \ln 8) - (\ln 2 + \tan^{-1} 0 - \frac{1}{2} \ln 4) \\
 &= \ln 4 + \frac{\pi}{4} - \frac{1}{2} \ln 8 - \ln 2 + \frac{1}{2} \ln 4 \\
 &= 2 \ln 2 + \frac{\pi}{4} - \frac{3}{2} \ln 2 - \ln 2 + \ln 2 \\
 &= \frac{1}{2} \ln 2 + \frac{\pi}{4}
 \end{aligned}$$

d). Parametric equations: $x = 3 + 2\lambda$
 $y = 2 - \lambda$
 $z = -1 - 2\lambda$

$$\begin{aligned}
 (x-3)^2 + (y-2)^2 + (z+1)^2 &= 36 \\
 (3+2\lambda-3)^2 + (2-\lambda-2)^2 + (-1-2\lambda+1)^2 &= 36 \\
 4\lambda^2 + \lambda^2 + 4\lambda^2 &= 36 \\
 9\lambda^2 &= 36 \\
 \lambda^2 &= 4 \\
 \lambda &= \pm 2
 \end{aligned}$$

$$\begin{aligned}
 \lambda = -2 : x &= 3 - 4 = -1 \\
 y &= 2 + 2 = 4 \\
 z &= -1 + 4 = 3 \quad (-1, 4, 3)
 \end{aligned}$$

$$\begin{aligned}
 \lambda = 2 : x &= 3 + 4 = 7 \\
 y &= 2 - 2 = 0 \\
 z &= -1 - 4 = -5 \quad (7, 0, -5)
 \end{aligned}$$

Points of intersection are
 $(-1, 4, 3)$ and $(7, 0, -5)$.

$$(12) e) (i) z_1 = 2 \operatorname{cis} \frac{4\pi}{5}, z_2 = 2 \operatorname{cis} \frac{7\pi}{15}$$

$$|z_1| = 2$$

$$|z_2| = 2$$

$$= |z_1|, \text{ i.e. } OA = OB.$$

$$\angle AOB = \arg z_1 - \arg z_2 = \frac{\pi}{3}$$

$\angle ABO$ (opposite OA) and $\angle BAO$ (opposite OB) are equal since angles opposite equal sides are equal.

$$\angle ABO = \angle BAO$$

$$= (\pi - \angle AOB) \div 2$$

$$= \frac{\pi}{3}$$

$$\therefore \angle ABO = \angle BAO = \angle AOB \text{ and}$$

$\triangle OAB$ is equilateral.

$$(ii) z_2 - z_1 = \vec{AB}$$

$\vec{AB}$ is $\vec{OB}$ rotated by $\frac{\pi}{3}$ clockwise.

$$z_2 - z_1 = z_2 \operatorname{cis} \left(-\frac{\pi}{3}\right)$$

$$= 2 \operatorname{cis} \left(\frac{7\pi}{15} - \frac{\pi}{3}\right)$$

$$= 2 \operatorname{cis} \frac{2\pi}{15}$$

⑬ (a) $v^2 = 6 - x - x^2$ m/s.

(i) $v = 0$, displacement is max.

$$0 = 6 - x - x^2$$

$$0 = (3+x)(2-x)$$

$$x = -3, 2.$$

Particle oscillates between
-3m and 2m.

(ii) $\frac{-3+2}{2} = -\frac{1}{2}$ m.

(iii) max. speed @ $x = -\frac{1}{2}$ (centre).

$$v^2 = 6 - \left(-\frac{1}{2}\right) - \left(-\frac{1}{2}\right)^2$$

$$= \frac{25}{4}$$

$$v = \pm \frac{5}{2} \text{ m/s}$$

Maximum speed is at 2.5 m/s.

(iv) $v^2 = 6 - x - x^2$
 $\frac{1}{2}v^2 = 3 - \frac{x}{2} - \frac{x^2}{2}$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

$$= \frac{d}{dx} \left(3 - \frac{x}{2} - \frac{x^2}{2} \right)$$

$$= -\frac{1}{2} - \frac{1}{2}(2x)$$

$$\therefore \ddot{x} = -\frac{1}{2} - x.$$

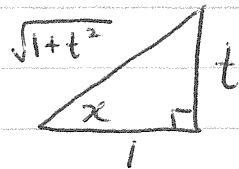
$$(13) (b) \int \frac{dx}{1 + \sin 2x}$$

$$t = \tan x$$

$$dt = \sec^2 x dx$$

$$= (1+t^2) dx$$

$$dx = \frac{dt}{1+t^2}$$



$$= \int \frac{1}{1 + 2\sin x \cos x} dx$$

$$= \int \frac{1}{1 + 2 \times \frac{t}{\sqrt{1+t^2}} \times \frac{1}{\sqrt{1+t^2}}} \frac{dt}{1+t^2}$$

$$= \int \frac{1}{1 + \frac{2t}{1+t^2}} \frac{dt}{1+t^2}$$

$$= \int \frac{1+t^2}{1+t^2+2t} \frac{dt}{1+t^2}$$

$$= \int \frac{1}{(t+1)^2} dt$$

$$= \int (t+1)^{-2} dt$$

$$= \frac{(t+1)^{-1}}{-1} + C$$

$$= -\frac{1}{t+1} + C$$

$$= -\frac{1}{\tan x + 1} + C$$

⑬(c). $\angle ABC$ found using $\underline{\underline{b}} \cdot \underline{\underline{c}} = |\underline{\underline{b}}| \cdot |\underline{\underline{c}}| \cos \theta$.
If $\cos 90^\circ = 0$, then $\underline{\underline{b}} \cdot \underline{\underline{c}} = 0$ and $\angle ABC = 90^\circ$
$$\begin{aligned}\underline{\underline{b}} \cdot \underline{\underline{c}} &= 1 \times 4 \times 8 \times (-2) + 3 \times 4 \\ &= 4 - 16 + 12 \\ &= 0.\end{aligned}$$

$\therefore \angle ABC = 90^\circ$ and $\triangle ABC$ is a right-angled triangle.

$$(13) (d). \quad I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx.$$

$$\begin{aligned}
 (i) \quad I_n &= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \times \sin x \, dx \\
 &= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \times \frac{d}{dx}(-\cos x) \, dx. \\
 &= \left[\sin^{n-1} x \times (-\cos x) \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x) \times \frac{d}{dx}(\sin^{n-1} x) \, dx \\
 &= 0 + \int_0^{\frac{\pi}{2}} \cos x \left[(n-1) \sin^{n-2} x \cos x \right] \, dx. \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx.
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad I_n &= (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx. \\
 &= (n-1) \int \sin^{n-2} x - \sin^n x \, dx. \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x \, dx \\
 &= (n-1) I_{n-2} - (n-1) I_n.
 \end{aligned}$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{n-1}{n} I_{n-2}.$$

$$\begin{aligned}
 (iii) \quad I_4 &= \frac{3}{4} I_2 \\
 I_2 &= \frac{1}{2} I_0 \\
 I_0 &= \int_0^{\frac{\pi}{2}} \sin^0 x \, dx \\
 &= \left[x \right]_0^{\frac{\pi}{2}} \\
 &= \frac{\pi}{2}
 \end{aligned}
 \quad \left\{ \begin{aligned}
 I_2 &= \frac{1}{2} \times \frac{\pi}{2} \\
 &= \frac{\pi}{4} \\
 I_4 &= \frac{3}{4} \times \frac{\pi}{4} \\
 &= \frac{3\pi}{16}.
 \end{aligned} \right.$$

$$(14) (a) \quad 4\ddot{x} + 6\ddot{x} = (6g - T) - (4g - T)$$

$$10\ddot{x} = 2g$$

$$\ddot{x} = \frac{g}{5}$$

$$v \cdot \frac{dv}{dx} = \frac{g}{5}$$

$$\frac{dv}{dx} = \frac{g}{5v}$$

$$\frac{dx}{dv} = \frac{5}{g} v$$

$$x = \frac{5}{g} \int_0^v v \, dv$$

$$= \frac{5}{g} \left[\frac{v^2}{2} \right]_0^v$$

$$x = \frac{5v^2}{2g}$$

$$\frac{2gx}{5} = v^2$$

$$v = \pm \sqrt{\frac{2gx}{5}}$$

$$v = \sqrt{\frac{2gx}{5}} \text{ only as Box } m \text{ is moving upwards.}$$



⑭ (b) Let $P(n)$ represent the proposition.

Show $P(n)$ true for $n=1$, i.e.

$$\frac{d}{dx}(xe^{2x}) = 2^{1-1}(2x+1)e^{2x}.$$

$$\text{LHS} = \frac{d}{dx}(xe^{2x})$$

$$= e^{2x} + x \cdot 2e^{2x}$$

$$= (2x+1)e^{2x}$$

$\therefore$ True for $n=1$.

$$\text{RHS} = 2^0(2x+1)e^{2x}$$

$$= (2x+1)e^{2x}$$

$$= \text{LHS}$$

Assume $P(n)$ true for $n=k$, i.e.

$$\frac{d^k}{dx^k}(xe^{2x}) = 2^{k-1}(2x+k)e^{2x}.$$

Show true for $n=k+1$, i.e.

$$\frac{d^{k+1}}{dx^{k+1}}(xe^{2x}) = 2^{k+1-1}(2x+k+1)e^{2x}.$$

$$= 2^k(2x+k+1)e^{2x}.$$

$$\text{LHS} = \frac{d^{k+1}}{dx^{k+1}}(xe^{2x})$$

$$= \frac{d}{dx} \left(\frac{d^k}{dx^k}(xe^{2x}) \right)$$

$$= \frac{d}{dx} \left(2^{k-1}(2x+k)e^{2x} \right) \quad \text{using assumption.}$$

$$= 2^{k-1} \times \left[(2x+k) \times 2e^{2x} + 2xe^{2x} \right]$$

$$= 2^{k-1} \times 2e^{2x} (2x+k+1)$$

$$= 2^{k-1} \times 2 \times (2x+k+1) \times e^{2x}$$

$$= 2^{k-1+1} (2x+k+1)e^{2x}$$

$$= 2^k (2x+k+1)e^{2x}$$

$$= \text{RHS}$$

14

(c)(i) $\ddot{x} = g - kv^2$

(ii) When $\ddot{x} = 0$, $g = kv^2$

$$v^2 = \frac{g}{k}$$

$$v = \sqrt{\frac{g}{k}}$$

(iii) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g - kv^2$$

$$\frac{d}{dv} \left(\frac{1}{2} v^2 \right) \frac{dv}{dx} = g - kv^2$$

$$v \frac{dv}{dx} = g - kv^2$$

$$\frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$x = -\frac{1}{2k} \int \frac{2kv}{g - kv^2} dv$$

$$= -\frac{1}{2k} \ln(g - kv^2) + C$$

$x=0, v=0$

$$0 = -\frac{1}{2k} \ln(g) + C$$

$$C = \frac{1}{2k} \ln g$$

$$x = -\frac{1}{2k} \ln(g - kv^2) + \frac{1}{2k} \ln g$$

$$= \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$$

⑭(d) Assume $\sqrt{a} + \sqrt{a+2} \leq \sqrt{a+8}$

$$(\sqrt{a} + \sqrt{a+2})^2 \leq a+8.$$

$$a + 2\sqrt{a}\sqrt{a+2} + a+2 \leq a+8.$$

$$2a + 2\sqrt{a(a+2)} + 2 \leq a+8$$

$$a + 2\sqrt{a(a+2)} \leq 6.$$

When $a=2$, $2+2+2\sqrt{2(2+2)} > 6$.

$$2+2\sqrt{2(2+2)} < 3+2\sqrt{3(3+2)}$$

$\therefore$ assumption is false.

⑮ (a) (i) $1^2 + 2^2 + 3^2 = 1 + 4 + 9$
 $= 14$

Also note that $4^2 > 14$, so
 a, b and c must be less than 4.

(ii) (x)
 centre = $\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$ and radius = $\sqrt{14}$.

(β) We require distance from centre
 to the coordinates to be $\sqrt{14}$ and
 of the form

$$\begin{pmatrix} 2 \pm a \\ 2 \pm b \\ 2 \pm c \end{pmatrix}$$

where $a^2 + b^2 + c^2 = 14$, satisfying
 conditions from part (i).

The permutations of $\pm 1, \pm 2, \pm 3$
 is $3! \times \underbrace{2 \times 2 \times 2}_{\substack{\uparrow \\ \text{is the permutations} \\ \text{of } 1, 2, 3.}} = 48$.

each ordinate
 can be positive
 or negative.

$$\textcircled{15} \text{ (b) (i) LHS} = \int_{\frac{1}{a}}^a \frac{f(x)}{x(f(x)+f(\frac{1}{x}))} dx. \quad u = \frac{1}{x} \quad \frac{du}{dx} = -\frac{1}{x^2} = -u^2$$

$$= \int_a^{\frac{1}{a}} \frac{f(\frac{1}{u})}{\frac{1}{u}[f(\frac{1}{u})+f(u)]} \times \left(-\frac{du}{u^2}\right)$$

$$= \int_{\frac{1}{a}}^a \frac{f(\frac{1}{u})}{u[f(u)+f(\frac{1}{u})]} du.$$

$$= \int_{\frac{1}{a}}^a \frac{f(\frac{1}{x})}{x[f(x)+f(\frac{1}{x})]} dx$$

$$= \text{RHS}$$

(ii)

$$I = \int_{\frac{1}{2}}^2 \frac{\sin x}{x(\sin x + \sin \frac{1}{x})} dx.$$

$$= \int_{\frac{1}{2}}^2 \frac{\sin(\frac{1}{x})}{x(\sin x + \sin \frac{1}{x})} dx.$$

$$2I = \int_{\frac{1}{2}}^2 \frac{\sin x + \sin \frac{1}{x}}{x(\sin x + \sin \frac{1}{x})} dx.$$

$$I = \frac{1}{2} \int_{\frac{1}{2}}^2 \frac{1}{x} dx.$$

$$= \frac{1}{2} [\ln x]_{\frac{1}{2}}^2$$

$$= \frac{1}{2} (\ln 2 - \ln \frac{1}{2})$$

$$= \frac{1}{2} (\ln 2 + \ln 2)$$

$$= \ln 2.$$

$$Q15(c)(i) \omega^n = 1, \quad 1 + \omega + \omega^2 + \omega^3 + \dots + \omega^{n-1} = 0$$

$$(1 + 2\omega + 3\omega^2 + 4\omega^3 + \dots + n\omega^{n-1})(\omega - 1)$$

$$= \omega + 2\omega^2 + 3\omega^3 + 4\omega^4 + \dots + (n-1)\omega^{n-1} + n\omega^n$$

$$- 1 - 2\omega - 3\omega^2 - 4\omega^3 - 5\omega^4 - \dots - n\omega^{n-1}$$

$$= -1 - \omega - \omega^2 - \omega^3 - \omega^4 - \dots - \omega^{n-1} + n\omega^n$$

$$= - (1 + \omega + \omega^2 + \omega^3 + \omega^4 + \dots + \omega^{n-1}) + n$$

$$= n \quad (\text{since } 1 + \omega + \omega^2 + \omega^3 + \dots + \omega^{n-1} = 0)$$

$$(ii) \text{ Let } z = cis \theta$$

$$= \cos \theta + i \sin \theta$$

$$\bar{z} = \cos \theta - i \sin \theta$$

$$z - \bar{z} = 2i \sin \theta$$

$$z^2 - 1 = cis 2\theta - 1$$

$$= \cos 2\theta + i \sin 2\theta - 1$$

$$\frac{1}{\cos 2\theta + i \sin 2\theta - 1} = \frac{1}{z^2 - 1}$$

$$= \frac{z^{-1}}{z - z^{-1}}$$

$$= \frac{\bar{z}}{z - \bar{z}} \quad \text{since } |z| = 1$$

$$= \frac{\cos \theta - i \sin \theta}{2i \sin \theta}$$

If $\sin \theta = 1$, then

$$\begin{aligned} & \cos 2\theta + i \sin 2\theta - 1 \\ &= 1 - 2\sin^2 \theta + 2i \sin \theta \cos \theta - 1 \\ &= \sin \theta (-2\sin \theta + 2i \cos \theta) \\ &= 0. \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{1}{w-1} &= \frac{1}{\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} - 1} \\ &= \frac{\cos \frac{\pi}{n} - i \sin \frac{\pi}{n}}{2i \sin \frac{\pi}{n}} \end{aligned}$$

$$= -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi}{n}.$$

$$\therefore \operatorname{Re}\left(\frac{1}{w-1}\right) = -\frac{1}{2}.$$

$$\text{(iv)} \quad 1 + 2w + 3w^2 + 4w^3 + \dots + nw^{n-1} = \frac{n}{w-1}$$

$$n=5: 1 + 2w + 3w^2 + 4w^3 + 5w^4 = \frac{5}{w-1} \quad (1)$$

$$\text{where } w = \operatorname{cis} \frac{2\pi}{5}$$

$$\operatorname{Re}(w^r) = \operatorname{Re}\left[\operatorname{cis} \frac{2\pi}{5}\right]^r$$

$$= \operatorname{Re}\left(\cos \frac{2\pi r}{5} + i \sin \frac{2\pi r}{5}\right)$$

$$= \cos \frac{2\pi r}{5} \quad (2)$$

$$\operatorname{Re}(1 + 2w + 3w^2 + 4w^3 + 5w^4) = \operatorname{Re}\left(\frac{5}{w-1}\right)$$

$$1 + 2\operatorname{Re}(w) + 3\operatorname{Re}(w^2) + 4\operatorname{Re}(w^3) + 5\operatorname{Re}(w^4) = 5\operatorname{Re}\left(\frac{1}{w-1}\right)$$

$$1 + 2\cos \frac{2\pi}{5} + 3\cos \frac{4\pi}{5} + 4\cos \frac{6\pi}{5} + 5\cos \frac{8\pi}{5} = -5 \times \frac{1}{2} = -\frac{5}{2}$$

(16) (a). $x^2 - 2x + 4 = 0$, α, β roots

(i) $x = \frac{2 \pm \sqrt{4-16}}{2}$

$$= \frac{2 \pm 2\sqrt{3}i}{2}$$

$$= 1 \pm \sqrt{3}i$$

$$\alpha = 1 + \sqrt{3}i, \quad \beta = 1 - \sqrt{3}i$$

$$|\alpha| = \sqrt{1+3} = 2, \quad \alpha = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}.$$

$$\alpha = 1 + \sqrt{3}i = 2 \operatorname{cis} \frac{\pi}{3}.$$

$$\alpha^n = 2^n \operatorname{cis} \frac{n\pi}{3}.$$

Similarly, $\beta^n = 1 - \sqrt{3}i$

$$= 2^n \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^n$$
$$= 2^n \left(\cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} \right).$$

$$\begin{aligned} \alpha^n - \beta^n &= 2^n \cos \frac{n\pi}{3} + 2^n i \sin \frac{n\pi}{3} \\ &\quad - 2^n \cos \frac{n\pi}{3} + 2^n i \sin \frac{n\pi}{3} \\ &= 2^n \left(2i \sin \frac{n\pi}{3} \right) \\ &= 2^{n+1} i \sin \frac{n\pi}{3}. \end{aligned}$$

(ii) $\alpha^9 - \beta^9 = 2^{10} i \sin \frac{9\pi}{3}$

$$= 2^{10} i \sin 3\pi$$
$$= 0.$$

16 (b)

(i) $z = \text{cis } \theta$

$$z^4 = \text{cis } 4\theta$$

$$= \sum_{k=0}^4 \binom{4}{k} i^k \sin^k \theta \cos^{4-k} \theta$$

$$\cos 4\theta = \binom{4}{0} \cos^4 \theta + \binom{4}{2} (-\sin^2 \theta) \cos^2 \theta + \binom{4}{4} \sin^4 \theta$$

$$\sin 4\theta = \binom{4}{1} \sin \theta \cos^3 \theta + \binom{4}{3} (-\sin^3 \theta) \cos \theta$$

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

$$= \frac{4 \sin \theta \cos^3 \theta - 4 \sin^3 \theta \cos \theta}{\cos^4 \theta - 6 \sin^2 \theta \cos^2 \theta + \sin^4 \theta} \quad \begin{matrix} \div \cos^4 \theta \\ \div \cos^4 \theta \end{matrix}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

(ii) $\tan 4\theta = 1$

$$4\theta = \tan^{-1}(1) + \pi n, \quad n \in \mathbb{J}$$

$$= \frac{\pi}{4} + \pi n, \quad n = 0, \pm 1, \pm 2, \dots$$

$$\theta = \frac{\pi}{16} + \frac{\pi n}{4}$$

$$= \frac{\pi(1+4n)}{16}, \quad n \in \mathbb{J}$$

(iii) $\frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} = 1 \quad (\text{let } \tan \theta = x)$

$$4 \tan \theta - 4 \tan^3 \theta = 1 - 6 \tan^2 \theta + \tan^4 \theta$$

$$4x - 4x^3 = 1 - 6x^2 + x^4$$

$$x^4 + 4x^3 - 6x^2 + 4x + 1 = 0$$

$$x = \tan \left[\frac{\pi}{16} (1+4n) \right], \quad n \in \mathbb{J}$$

$$= \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{3\pi}{16}, -\tan \frac{7\pi}{16}$$

(16) c)(i) Given $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$

$$\frac{t}{3} \geq (abc)^{\frac{1}{3}}$$

$$\frac{3}{t} \leq \frac{1}{(abc)^{\frac{1}{3}}}$$

$$\frac{1}{(abc)^{\frac{1}{3}}} \geq \frac{3}{t} \quad \star$$

$$\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \sqrt[3]{\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3 \left(\frac{1}{abc} \right)^{\frac{1}{3}}$$

$$\geq 3 \left(\frac{3}{t} \right) \quad \text{using } \star$$

$$\geq \frac{9}{t}$$

$$(ii) \left(\frac{1}{a} - 1 \right) \left(\frac{1}{b} - 1 \right) \left(\frac{1}{c} - 1 \right) = \left(\frac{1-a}{a} \right) \left(\frac{1-b}{b} \right) \left(\frac{1-c}{c} \right)$$

$$= \frac{(1-a)(1-b)(1-c)}{abc}$$

$$= \frac{1 - (a+b+c) + (ab+ac+bc) - abc}{abc}$$

$$= \frac{1 - 1 + ab+ac+bc - abc}{abc}$$

$$= \frac{1}{c} + \frac{1}{b} + \frac{1}{a} - 1$$

$$\geq \frac{9}{t} - 1$$

$$\geq \frac{9}{a+b+c} - 1$$

$$\geq \frac{9}{1} - 1$$

$$\geq 8$$